

Arunav Bordoloi

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Department of Physics and Center for Quantum Information Physics
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New York University

Employment:

New York University (NYU)

Postdoctoral Associate, March 2025 – present
Advisor: Javad Shabani

University of Maryland (UMD), College Park

Postdoctoral Associate, March 2022 – February 2025
Advisor: Vladimir Manucharyan

Education:

University of Basel, Switzerland

Ph.D. in Physics, September 2016 – January 2021
Bridging Postdoc, February 2021 – December 2021
Spin Projection and Correlation Experiments in Nanoelectronic Devices
Advisor: Christian Schoenenberger

Indian Institute of Science Education and Research Kolkata

B.S. and M.S. in Physical Sciences, August 2011 – June 2016
Microwave Cavity Perturbation on $\text{NH}_4\text{CuPO}_4 \cdot \text{H}_2\text{O}$ single crystals and YBCO Thin Films at Low Temperature
Advisor: Chiranjib Mitra

Research Interests:

Hybrid superconducting–semiconducting quantum systems and devices;
Superconducting qubits and quantum information science; Circuit quantum electrodynamics; Mesoscopic quantum transport and phase transitions;
Cooper pair splitting in quantum dot architectures etc.

Honors and Awards:

- *Material Physics: Highlights of 2023*. Institute of Physics (IOP), Physics World, 2023.
 - *Swiss Nanotechnology PhD Award*, 2023.
 - *Wilhelm and Else Heraeus-Stiftung Prize* (Best Poster). Bad Honnef, Germany, 2019.
 - *Charpak Scholarship*. Government of France, 2014.
 - *Summer Research Fellowship*. Indian Academy of Sciences, 2013.
 - *INSPIRE Fellowship*. Department of Science and Technology, India, 2011 – 2016.
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Publications: (*equal contribution, †corresponding author)

- **A. Bordoloi**[†], V. Zannier, L. Sorba, C. Schönenberger, A. Baumgartner. Spin cross-correlation experiments in an electron entangler. *Nature* 612, 454–458 (2022).
- S. Liu*, **A. Bordoloi**^{*,†}, J. Issokson, I. Levy, M.G. Vavilov, J. Shabani, V.E. Manucharyan. Strongly anharmonic flux-tunable transmon based on InAs-Al 2D

heterostructure.

Nature Communications 17, 740 (2025).

- **A. Bordoloi**[†], V. Zannier, L. Sorba, C. Schönenberger, A. Baumgartner.
A double quantum dot spin valve.
Communication Physics 3, 135 (2020).
- M. Mikalsen, A.G. Penner, S.D. Escribano, N. Drechsler, **A. Bordoloi**, J. Issokson, F.V. Oppen, Y. Oreg, J. Shabani.
Observation of Critical Current Minimum in Super-Honeycomb Josephson Junction Arrays.
arXiv:2607.16500. *in review, Physical Review Letters* (2026).
- F.H. Knudsen, A. Leblanc, P.J. Strohbeen, J. van Dijk, Y. Li, L. Kusher, **A. Bordoloi**, A. Danilenko, X. Ma, S. Salmani-Rezaie, J. Shabani.
Crystalline Group-IV Josephson Junction.
arXiv:2608.08500. *in review, Nano Letters* (2026).
- L.S. Kusher, D. Peng, Z. Zhu, **A. Bordoloi**, A. Leblanc, L.J. Baker, N. Adnan, J. Issokson, A. Wang, F. Knudsen, K. Dindial, M. Mikalsen, T. Kaleem, A. Vrajitoarea, Y. Du, P.J. Strohbeen, J. Shabani.
Effect of Buried-Interface Preparation for Nb Superconducting Resonators on InP.
arXiv:2609.13673. *in review, ACS Applied Materials & Interfaces* (2026).
- S. Liu, **A. Bordoloi**, M.G. Vavilov, J. Shabani, J. Issokson, V.E. Manucharyan, K. Sardashti.
Qubit spectroscopy of the Fraunhofer-modulated modes in InAs 2DEG Josephson weak links. (in preparation)

See also: google scholar

Invited Talks:

- “Strongly anharmonic flux-tunable transmon based on InAs-Al two dimensional heterostructure.” Quantum Materials Seminar.
Columbia University, New York, USA (September 2026).
- “Strongly anharmonic flux-tunable transmon based on InAs-Al two dimensional heterostructure.” R. G. Herb Condensed Matter Seminar.
University of Wisconsin-Madison, USA (September 2026).
- “Strongly anharmonic flux-tunable transmon based on InAs-Al two-dimensional heterostructure.” Center for Quantum Information Science and Engineering Seminar.
Virginia Tech, USA (April 2026).
- “Flux tunable qubits and spin cross-correlations experiments in superconductor - semiconductor heterostructures.” Department of Physics Seminar.
University of Chicago, USA (January 2025).
- “Flux tunable qubits and quantum correlations in superconductor - semiconductor hybrid heterostructures.” Center for Quantum Information Physics Seminar.
New York University (NYU), USA (October 2024).
- “Quantum Dot Spin Valves and Spin Correlation Experiments in Cooper Pair Splitter.” Department of Physics Seminar.
University of Rochester, USA (virtual) (October 2021).
- “Quantum Dot Spin Valves and Spin Correlation Experiments in Cooper

Pair Splitter.” Center for Quantum Information Physics Seminar.
New York University (NYU), USA (virtual) (September 2021).

- “Quantum Dot Spin Valves and Spin Correlation Experiments in Cooper Pair Splitter.” Department of Physics Seminar.
Institute of Science and Technology (IST), Vienna, Austria (August 2021).
 - “Quantum Dot Spin Valves and Spin Correlation Experiments in Cooper Pair Splitter.” Department of Physics Seminar.
Purdue University, USA (virtual) (August 2021).
 - “Quantum Dot Spin Valves and Spin Correlation Experiments in Cooper Pair Splitter.” Department of Physics Seminar.
University of Pittsburgh, USA (virtual) (July 2021).
 - “A Double Quantum Dot Spin Valve.” Young Research Leaders Group Workshop.
Mainz, Germany (November 2020).
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Contributed Talks:

- “Hybrid InAs–Al Josephson Architectures for Exploring Quantum Phase Transitions.” American Physical Society Global Physics Summit, Denver, USA (March 2026).
- “Gateless Gatemon Qubit in a Two-dimensional InAs–Al heterostructure.” American Physical Society Global Physics Summit, Anaheim, USA (March 2025).
- “Towards RF spectroscopy of Andreev bound states in an InAs–Al 2DEG.” American Physical Society March Meeting, Minneapolis, USA (March 2024).
- “Spin Cross-Correlation Experiments in Semiconducting– Superconducting Heterostructures.” Quantum Matter Conference, Madrid, Spain (May 2023).
- “A Double Quantum Dot Spin Valve.” Joint Quantum Institute (JQI) Seminar, University of Maryland (UMD), College Park, USA (April 2023).
- “Spin Cross-Correlation Experiments in a Cooper Pair Splitter.” Joint Quantum Institute (JQI) Seminar, University of Maryland (UMD), College Park, USA (March 2023).
- “Negative Spin Cross-Correlation in a Cooper Pair Splitter.” American Physical Society March Meeting, Chicago, USA (March 2022).
- “Negative Spin Current Correlation in a Cooper Pair Splitter.” Joint Annual Meeting of Swiss Physical Society and Austrian Physical Society, Innsbruck, Austria (June 2021).
- “Negative Spin Current Correlation in a Cooper Pair Splitter.” Institute of Physics (IOP) Bound States in Hybrid Superconductor Nanostructures Conference, (virtual), UK (April 2021).
- “Spin (Current) Control in Nanowire Devices with Ferromagnetic Sidegates (poster).” NCCR SPIN Annual Meeting, Swiss National Science Foundation, Pontresina, Switzerland (April 2021).
- “A Double Quantum Dot Spin Valve.” CMD2020GEFES, European Physical Society, UK (virtual) (August 2020).
- “A Double Quantum Dot Spin Valve.” QSIT Lunch Seminar, ETH Zurich,

Switzerland (May 2020).

- “Double Quantum Dot Tunneling Magnetoresistance without Ferromagnetic Contacts.” One Dimensional Systems for Quantum Technology Conference, Germany (July 2019).
 - “Towards Synthetic Spin-Orbit Interaction in InAs Nanowires using Ferromagnetic Sidegates.” “Transport in Nanostructures” Summer School, Capri, Italy (May 2017).
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**Teaching,
Mentorship and
Service:**

- **Research Mentor:**
 - Shukai Liu, PhD Candidate, UMD (2022–2025).
 - Jakob Issokson, PhD Candidate, NYU (2025–present).
 - Tongzhou Wang, PhD Student, NYU (2025–present)
 - Nichae Adnan, PhD Student, NYU (2025–present)
 - Leo Uhre Jakobsen, PhD Student, NYU (2025–present)
 - More than 10 undergraduate researchers (2022-present).
 - **Teaching:**
 - Physics 1 - Classical Mechanics, University of Basel, Switzerland (*Spring Semester (SS) 2016, SS 2017, SS 2018, SS2019, SS 2020*)
 - Quantum Transport in Nanostructures (*Fall Semester (FS) 2017*).
 - Experimental Physics (*FS 2018*).
 - Physics 2 - Electromagnetism (*FS 2019, FS 2021*).
 - **Reviewer:** Nature Communications (1 manuscript); Physical Review X (1 manuscript); Physical Review Applied (2 manuscripts); Academia Quantum (1 manuscript).
 - **Service and Outreach:**
 - Program Coordinator, Summer Quantum Camp, New York City (July-August 2026)
 - Lead Instructor, Summer Quantum Camp, New York City (July-August 2025)
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Techniques:

- **Micro and Nanofabrication:** Electron Beam Lithography (EBL), Scanning Electron Microscopy (SEM), Reactive Ion Etching (RIE/ICP), Wet and Dry Chemical Etching, Sputtering, E-beam Evaporation.
- **Computational Modeling and Simulation:** ANSYS HFSS, ANSYS Maxwell, COMSOL Multiphysics, Sonnet.
- **Design and Layout Automation:** KLayout, Qubit Design, Autodesk Inventor, Autodesk AutoCAD.
- **Cryogenic and Vacuum Systems:** mK Dilution Refrigerators, Ultra-High Vacuum (UHV) Systems.
- **Instrumentation and RF Spectroscopy:** Microwave/RF Spectroscopy, QCoDeS, Labber, QUA.
- **Programming and Data Analysis:** Python, MATLAB, C, Fortran 90.
- **Scientific Vector Graphics:** Adobe Illustrator, Inkscape, Adobe Photoshop.

References:

Prof. Javad Shabani

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and Computer Engineering
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formation Physics
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sanne (EPFL)
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Prof. Maxim Vavilov

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